14_evaluation_detection_wGenes

August 31, 2026

[1]:
```
library(Seurat)
# library(SeuratDisk)

library(reticulate)
library(anndata)

library(ggplot2)
library(ggpubr)
library(pheatmap)
library(dplyr)
library(tidyr)
library(RColorBrewer)
library(clustree)
library(repr)
library(ggdist)
library (ggridges)
options(repr.plot.width=10, repr.plot.height=8)


library(UpSetR)
library(grid)

library(PRROC)
library(Matrix)

library(reshape2)

library(GenomicRanges)
library(GenomicFeatures)

library(purrr)

library(scales)

getwd()

dataset_id <- "simulated_mm_RA_wGenes"
```

```r
genome_id <- "mm10"
samples <- c("all")

dir.create("figures")
for (sample in samples) {
    dir.create(paste0("figures/figures_", dataset_id, "_", sample))
}
dir.create(paste0("figures/figures_", dataset_id))


colorTools <- c( # "MATES"="#F98A7B",
    "STARsolo_TE" = "#FFBC81",
    "STARsolo_TE_EM" = "#F3E088",
    "SoloTE_unique" = "#A4DD9B",
    "SoloTE_thr2" = "#6FC69D",
    "SoloTE_thr1" = "#4BB2BB",
    "SoloTE_thr0" = "#3989BF",
    "Stellarscope" = "#4F5D93",
    "simulated" = "grey70"
)

thrMinCells <- 500 * 0.05
```

```
Loading required package: SeuratObject

Loading required package: sp


Attaching package: 'SeuratObject'


The following objects are masked from 'package:base':

    intersect, t



Attaching package: 'anndata'


The following object is masked from 'package:SeuratObject':

    Layers



Attaching package: 'dplyr'
```

```
The following objects are masked from ‘package:stats’:

    filter, lag


The following objects are masked from ‘package:base’:

    intersect, setdiff, setequal, union


Loading required package: ggraph


Attaching package: ‘ggraph’


The following object is masked from ‘package:sp’:

    geometry


Attaching package: ‘ggridges’


The following objects are masked from ‘package:ggdist’:

    scale_point_color_continuous, scale_point_color_discrete,
    scale_point_colour_continuous, scale_point_colour_discrete,
    scale_point_fill_continuous, scale_point_fill_discrete,
    scale_point_size_continuous


Loading required package: rlang


Attaching package: ‘rlang’


The following object is masked from ‘package:ggdist’:

    ll


Attaching package: ‘Matrix’
```

```
The following objects are masked from ‘package:tidyr’:

    expand, pack, unpack


Attaching package: ‘reshape2’

The following object is masked from ‘package:tidyr’:

    smiths

Loading required package: stats4

Loading required package: BiocGenerics

Attaching package: ‘BiocGenerics’

The following objects are masked from ‘package:dplyr’:

    combine, intersect, setdiff, union

The following object is masked from ‘package:SeuratObject’:

    intersect

The following objects are masked from ‘package:stats’:

    IQR, mad, sd, var, xtabs

The following objects are masked from ‘package:base’:

    anyDuplicated, aperm, append, as.data.frame, basename, cbind,
    colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
    get, grep, grepl, intersect, is.unsorted, lapply, Map, mapply,
    match, mget, order, paste, pmax, pmax.int, pmin, pmin.int,
    Position, rank, rbind, Reduce, rownames, sapply, setdiff, sort,
    table, tapply, union, unique, unsplit, which.max, which.min
```

```
Loading required package: S4Vectors

Attaching package: ‘S4Vectors’

The following objects are masked from ‘package:Matrix’:

    expand, unname

The following object is masked from ‘package:tidyr’:

    expand

The following objects are masked from ‘package:dplyr’:

    first, rename

The following object is masked from ‘package:utils’:

    findMatches

The following objects are masked from ‘package:base’:

    expand.grid, I, unname

Loading required package: IRanges

Attaching package: ‘IRanges’

The following objects are masked from ‘package:dplyr’:

    collapse, desc, slice

The following object is masked from ‘package:sp’:

    %over%

Loading required package: GenomeInfoDb
```

```
Loading required package: AnnotationDbi

Loading required package: Biobase

Welcome to Bioconductor

    Vignettes contain introductory material; view with
    'browseVignettes()'. To cite Bioconductor, see
    'citation("Biobase")', and for packages 'citation("pkgname")'.



Attaching package: ‘Biobase’


The following object is masked from ‘package:rlang’:

    exprs



Attaching package: ‘AnnotationDbi’


The following object is masked from ‘package:dplyr’:

    select



Attaching package: ‘purrr’


The following object is masked from ‘package:GenomicRanges’:

    reduce


The following object is masked from ‘package:IRanges’:

    reduce


The following objects are masked from ‘package:rlang’:

    %@%, flatten, flatten_chr, flatten_dbl, flatten_int, flatten_lgl,
    flatten_raw, invoke, splice
```

```
Attaching package: ‘scales’


The following object is masked from ‘package:purrr’:

    discard
```

’/mnt/transfer/TEbenchmarking/Rscripts’

```
Warning message in dir.create("figures"):
"'figures' already exists"
Warning message in dir.create(paste0("figures/figures_", dataset_id, "_",
sample)):
"'figures/figures_simulated_mm_RA_wGenes_all' already exists"
Warning message in dir.create(paste0("figures/figures_", dataset_id)):
"'figures/figures_simulated_mm_RA_wGenes' already exists"
```

[ ]:
```
load("workspaces/13_evaluation_objectCreation_wGenes.Rdata")
```

1 Evaluation

1.1 Upset plots of detected TEs and TPs

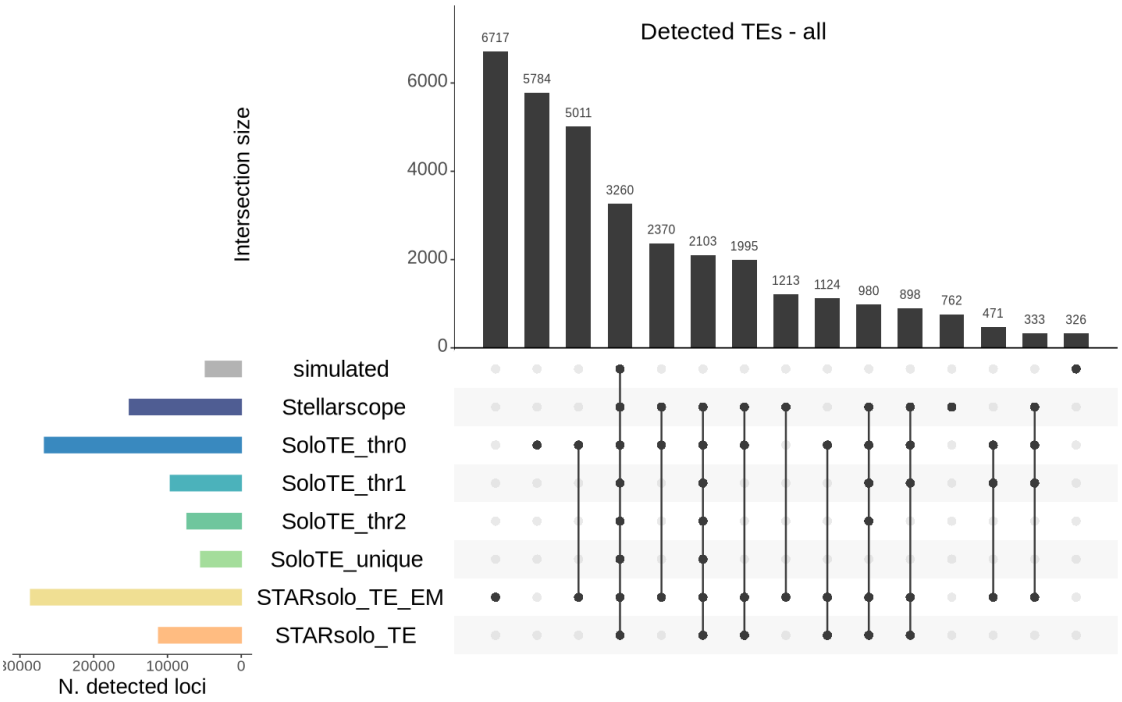
1.1.1 Upsets of detected TEs

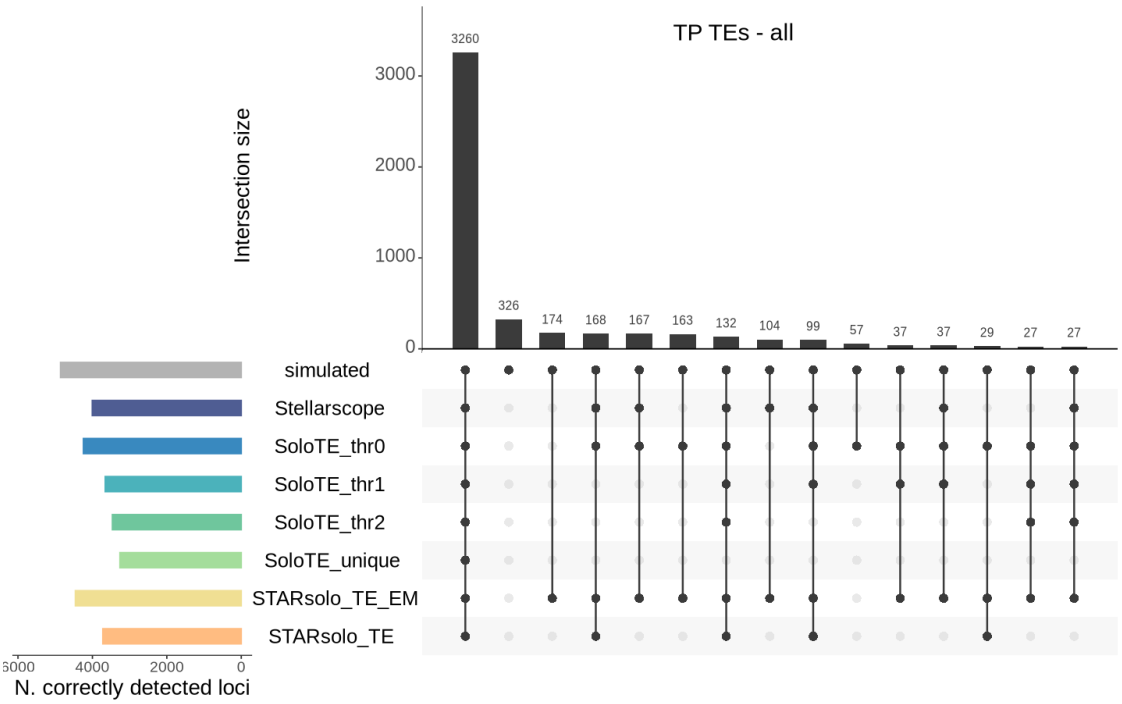
[3]:
```
options(repr.plot.width=12, repr.plot.height=7.6)

for (sample in samples){

    print(sample)

    TPlist <- list()
    FNlist <- list()
    FPlist <- list()

    detectedList <- list()

    for (tool in c(names(objList), "simulated")){

        print(tool)
        if (tool == "simulated") {
            obj <- splatter_objs[[sample]]
        }else {
            obj <- objList[[tool]][[sample]]
```

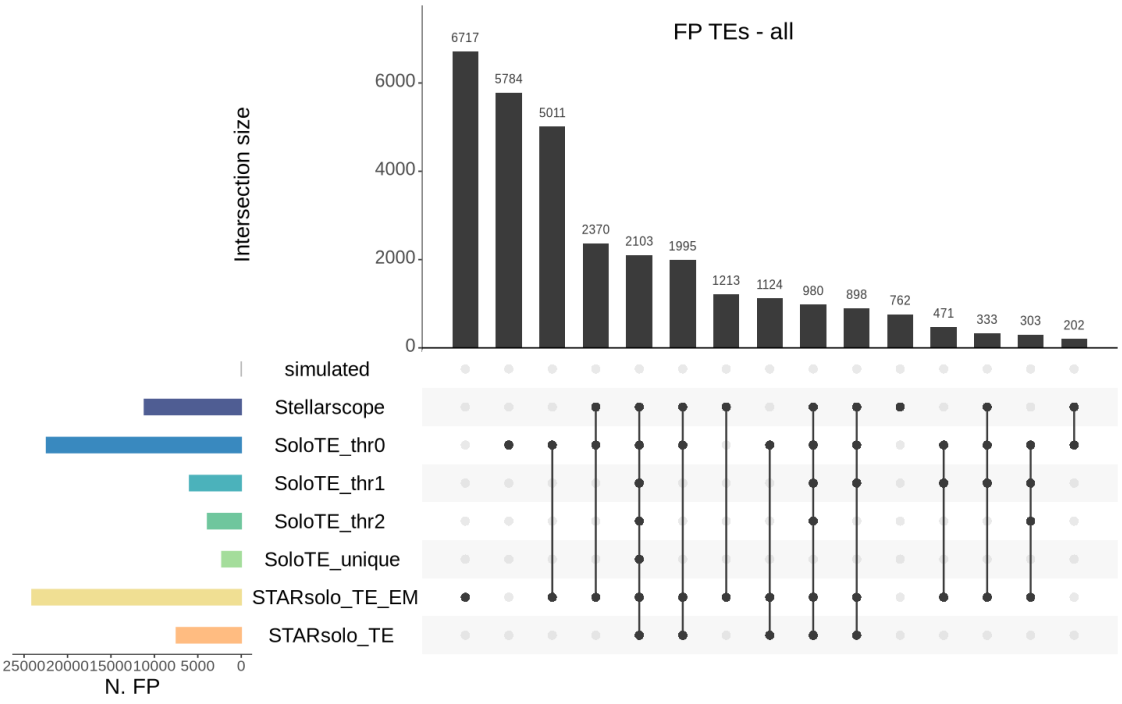
```
      }

      detectedList[[sub("^(.*)_.*$", "\\1", obj@project.name)]] <-␣
↪Features(obj)
      TPlist[[sub("^(.*)_.*$", "\\1", obj@project.name)]] <-␣
↪intersect(Features(splatter_objs[[sample]]), Features(obj))
      FNlist[[sub("^(.*)_.*$", "\\1", obj@project.name)]] <-␣
↪setdiff(Features(splatter_objs[[sample]]), Features(obj))
      FPlist[[sub("^(.*)_.*$", "\\1", obj@project.name)]] <-␣
↪setdiff(Features(obj), Features(splatter_objs[[sample]]))
  }

  #pdf(file=paste0("figures/figures_", dataset_id, "_",sample,"/
↪upset_detectedTEs.pdf"), width = 12, height = 7.6)
  show(UpSetR::upset(fromList(detectedList), nintersects = 15,
          sets=names(colorTools), keep.order = T, sets.bar.color=colorTools,
          text.scale = c(2, 2, 2, 1.65, 2.5, 1.5),
          order.by=c("freq"),
          point.size=3, mb.ratio = c(0.5, 0.5),
          set_size.show=F, set_size.scale_max=␣
↪max(lengths(detectedList))+1000, #show.numbers = F,
          nsets=length(objList)+1,
          sets.x.label="N. detected loci",
          mainbar.y.label="Intersection size")

  )
  grid.text(paste0("Detected TEs - ", sample ),x = 0.65, y=0.95,␣
↪gp=gpar(fontsize=18))
  #dev.off()

  #pdf(file=paste0("figures/figures_", dataset_id, "_",sample,"/
↪upset_detected_TP.pdf"), width = 11, height = 7.5)
  show(UpSetR::upset(fromList(TPlist), nintersects = 15,
          sets=names(colorTools), keep.order = T, sets.bar.color=colorTools,
          text.scale = c(2, 2, 2, 1.65, 2.2, 1.5),
          order.by=c("freq"),
          point.size=3, mb.ratio = c(0.5, 0.5),
          set_size.show=F, set_size.scale_max= max(lengths(TPlist))+1000,␣
↪#show.numbers = F,
          nsets=length(c(objList, splatter_obj)),
          sets.x.label="N. correctly detected loci",
          mainbar.y.label="Intersection size")
  )
  grid.text(paste0("TP TEs - ", sample ),x = 0.65, y=0.95,␣
↪gp=gpar(fontsize=18))
```

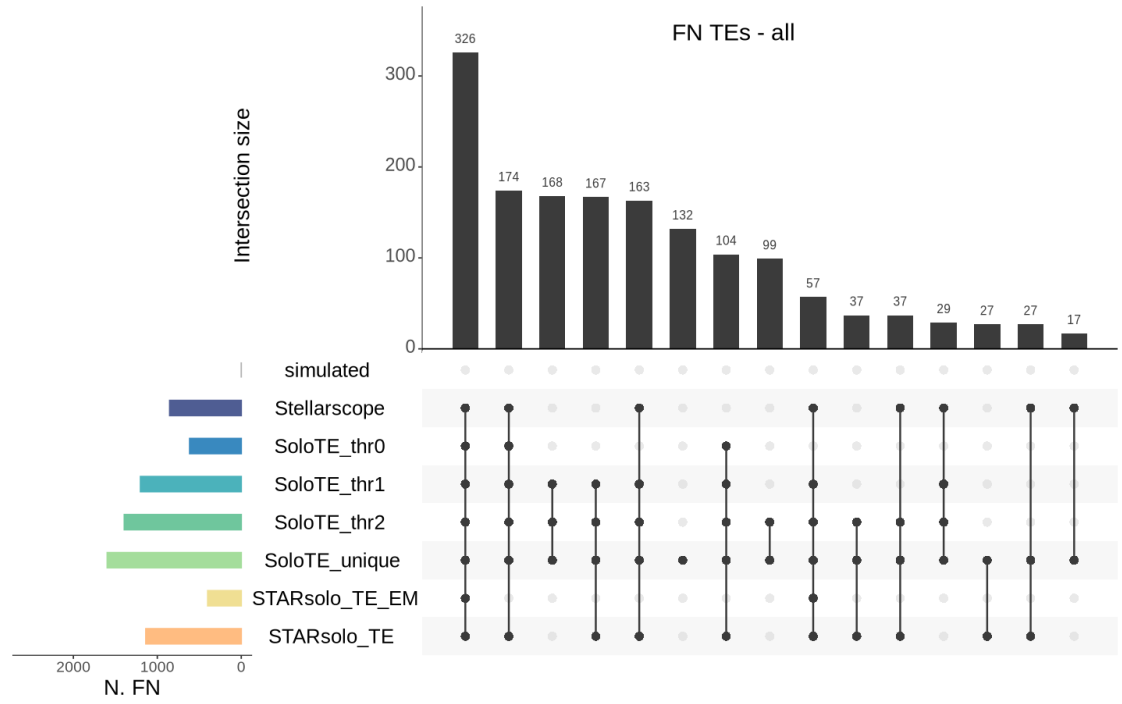
```
    show(UpSetR::upset(fromList(FPlist), nintersects = 15,
            sets=names(colorTools), keep.order = T, sets.bar.color=colorTools,
            text.scale = c(2, 2, 2, 1.65, 2.2, 1.5),
            order.by=c("freq"),
            point.size=3, mb.ratio = c(0.5, 0.5),
            set_size.show=F, set_size.scale_max= max(lengths(FPlist))+1000,␣
 ↪#show.numbers = F,
            nsets=length(c(objList, splatter_obj)),
            sets.x.label="N. FP",
            mainbar.y.label="Intersection size")
    )
    grid.text(paste0("FP TEs - ", sample ),x = 0.65, y=0.95,␣
 ↪gp=gpar(fontsize=18))

    show(UpSetR::upset(fromList(FNlist), nintersects = 15,
            sets=names(colorTools), keep.order = T, sets.bar.color=colorTools,
            text.scale = c(2, 2, 2, 1.65, 2.2, 1.5),
            order.by=c("freq"),
            point.size=3, mb.ratio = c(0.5, 0.5),
            set_size.show=F, set_size.scale_max= max(lengths(FNlist))+1000,␣
 ↪#show.numbers = F,
            nsets=length(c(objList, splatter_obj)),
            sets.x.label="N. FN",
            mainbar.y.label="Intersection size")
    )
    grid.text(paste0("FN TEs - ", sample ),x = 0.65, y=0.95,␣
 ↪gp=gpar(fontsize=18))
    #|dev.off()
}
```

```
[1] "all"
[1] "STARsolo_TE"
[1] "STARsolo_TE_EM"
[1] "SoloTE_unique"
[1] "SoloTE_thr2"
[1] "SoloTE_thr1"
[1] "SoloTE_thr0"
[1] "Stellarscope"
[1] "simulated"
```


Detected TEs - all
Intersection size
6717
5784
5011
3260
2370
2103
1995
1213
1124
980
898
762
471
333
326
6000
4000
2000
0
simulated
Stellarscope
SoloTE_thr0
SoloTE_thr1
SoloTE_thr2
SoloTE_unique
STARsolo_TE_EM
STARsolo_TE
30000
20000
10000
0
N. detected loci


TP TEs - all
Intersection size
3260
326
174
168
167
163
132
104
99
57
37
37
29
27
27
3000
2000
1000
0
simulated
Stellarscope
SoloTE_thr0
SoloTE_thr1
SoloTE_thr2
SoloTE_unique
STARsolo_TE_EM
STARsolo_TE
5000
4000
2000
0
N. correctly detected loci


FP TEs - all
Intersection size
6717
5784
5011
2370
2103
1995
1213
1124
980
898
762
471
333
303
202
6000
4000
2000
0
simulated
Stellarscope
SoloTE_thr0
SoloTE_thr1
SoloTE_thr2
SoloTE_unique
STARsolo_TE_EM
STARsolo_TE
25000 20000 15000 10000 5000 0
N. FP


FN TEs - all
Intersection size
326
174
168
167
163
132
104
99
57
37
37
29
27
27
17
300
200
100
0
simulated
Stellarscope
SoloTE_thr0
SoloTE_thr1
SoloTE_thr2
SoloTE_unique
STARsolo_TE_EM
STARsolo_TE
2000 1000 0
N. FN

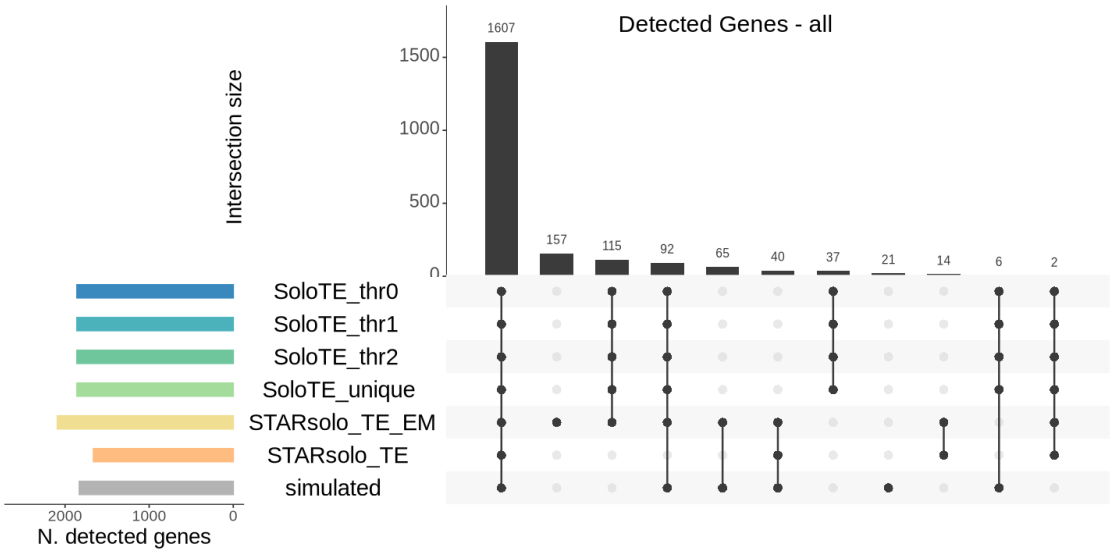
1.1.2 Upsets of detected Genes

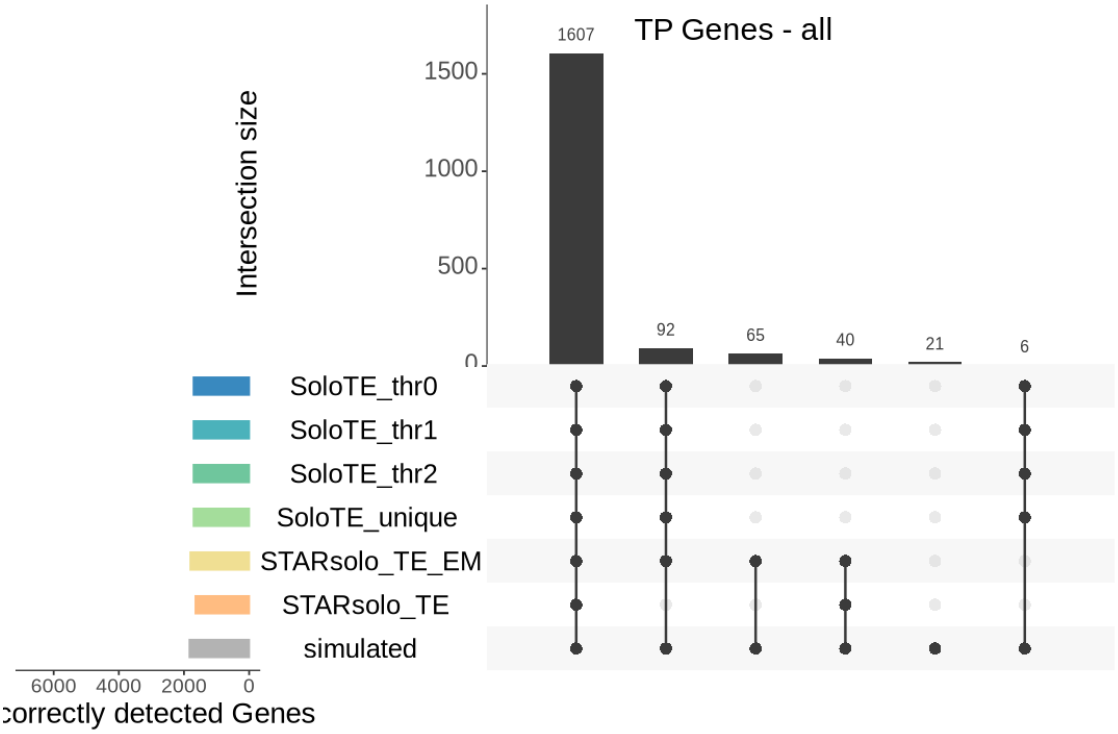
[4]:
```
options(repr.plot.width=12, repr.plot.height=7)

for (sample in samples){

    print(sample)

    TPgenelist <- list()
    FPgenelist <- list()
    FNgenelist <- list()
    detectedgeneList <- list()

    for (tool in c(names(objGeneList), "simulated")){

        print(tool)
        if (tool == "simulated") {
            obj <- splatter_objs_genes[[sample]]
        }else {
            obj <- objGeneList[[tool]][[sample]]
        }

        detectedgeneList[[sub("^(.*)_.*$", "\\1", obj@project.name)]] <-␣
↪Features(obj)
        TPgenelist[[sub("^(.*)_.*$", "\\1", obj@project.name)]] <-␣
↪intersect(Features(splatter_objs_genes[[sample]]), Features(obj))
        FNgenelist[[sub("^(.*)_.*$", "\\1", obj@project.name)]] <-␣
↪setdiff(Features(splatter_objs_genes[[sample]]), Features(obj))
        FPgenelist[[sub("^(.*)_.*$", "\\1", obj@project.name)]] <-␣
↪setdiff(Features(obj), Features(splatter_objs_genes[[sample]]))
    }

    options(repr.plot.width=12, repr.plot.height=6)
    # pdf(file=paste0("figures/figures", "_", dataset_id, "_",sample,"/
↪upset_detected.pdf"), width = 12, height = 6)
    show(UpSetR::upset(fromList(detectedgeneList), nintersects = 15,
            sets=names(colorTools[c("simulated", names(objGeneList))]), keep.
↪order = T, sets.bar.color=colorTools[c("simulated", names(objGeneList))],
            text.scale = c(2, 2, 2, 1.65, 2.5, 1.5),
            order.by=c("freq"),
            point.size=3, mb.ratio = c(0.5, 0.5),
            set_size.show=F, set_size.scale_max=␣
↪max(lengths(detectedgeneList))+500, #show.numbers = F,
            nsets=length(objGeneList)+1,
            sets.x.label="N. detected genes",
            mainbar.y.label="Intersection size")
```

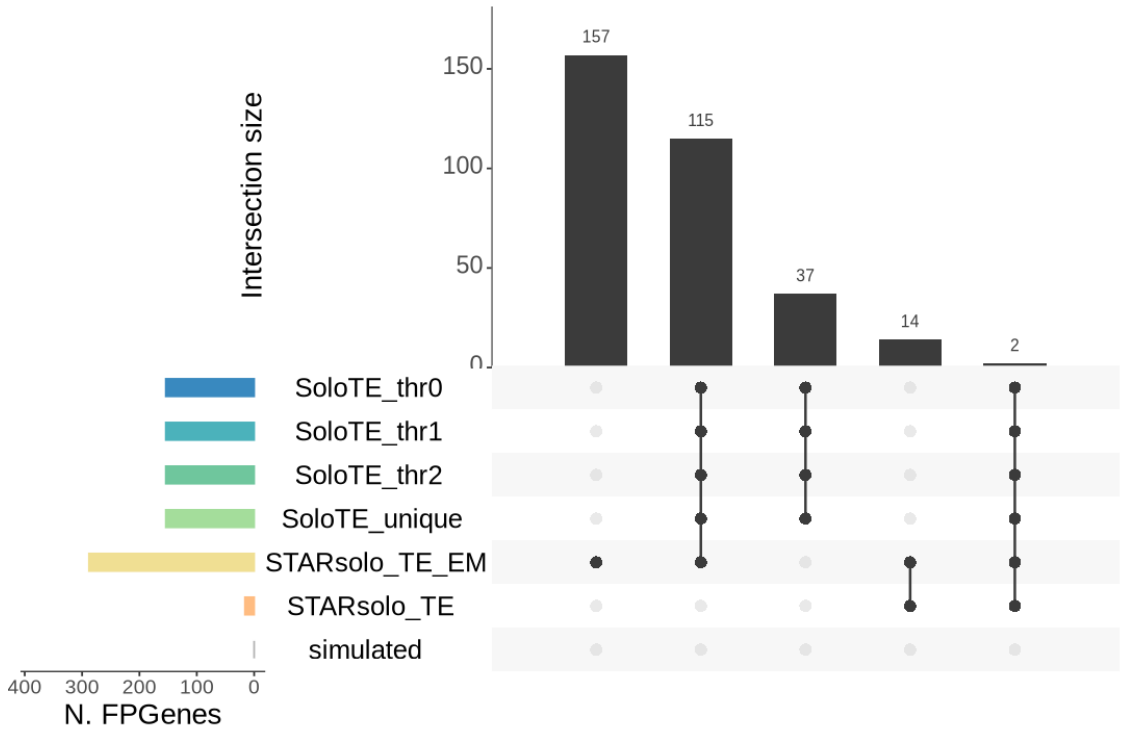
```
    )
    grid.text(paste0("Detected Genes - ", sample ),x = 0.65, y=0.95,␣
↪gp=gpar(fontsize=18))
    #dev.off()

    #pdf(file=paste0("figures/figures", "_", dataset_id, "_",sample,"/
↪upset_detected_TP.pdf"), width = 9, height = 6)
    show(UpSetR::upset(fromList(TPgenelist), nintersects = 15,
            sets=names(colorTools[c("simulated", names(objGeneList))]), keep.
↪order = T, sets.bar.color=colorTools[c("simulated", names(objGeneList))],
            text.scale = c(2, 2, 2, 1.65, 2.2, 1.5),
            order.by=c("freq"),
            point.size=3, mb.ratio = c(0.5, 0.5),
            set_size.show=F, set_size.scale_max= max(lengths(TPgenelist))+5000,␣
↪#show.numbers = F,
            nsets=length(c(objGeneList, splatter_obj)),
            sets.x.label="N. correctly detected Genes",
            mainbar.y.label="Intersection size")
    )
    grid.text(paste0("TP Genes - ", sample ),x = 0.65, y=0.95,␣
↪gp=gpar(fontsize=18))
    #dev.off()

    options(repr.plot.width=9, repr.plot.height=6)
    # pdf(file=paste0("figures/figures_", dataset_id, "_",sample,"/
↪upset_detected_TP.pdf"), width = 9, height = 6)
    show(UpSetR::upset(fromList(FPgenelist), nintersects = 15,
            sets=names(colorTools[c("simulated", names(objGeneList))]), keep.
↪order = T, sets.bar.color=colorTools[c("simulated", names(objGeneList))],
            text.scale = c(2, 2, 2, 1.65, 2.2, 1.5),
            order.by=c("freq"),
            point.size=3, mb.ratio = c(0.5, 0.5),
            set_size.show=F, set_size.scale_max= max(lengths(FPgenelist))+100,␣
↪#show.numbers = F,
            nsets=length(c(objGeneList, splatter_obj)),
            sets.x.label="N. FPGenes",
            mainbar.y.label="Intersection size")
    )
    # grid.text(paste0("FP Genes - ", sample ),x = 0.65, y=0.95,␣
↪gp=gpar(fontsize=18))
    # dev.off()
}
```

```
[1] "all"
[1] "STARsolo_TE"
[1] "STARsolo_TE_EM"
[1] "SoloTE_unique"
```

```
[1] "SoloTE_thr2"
[1] "SoloTE_thr1"
[1] "SoloTE_thr0"
[1] "simulated"
```







2 TE - gene intresection

[5]:
```
library(rtracklayer)

gene_annotation <- rtracklayer::import("/mnt/transfer/TEbenchmarking/Rscripts/
  ↪annotation/gencode.vM10.primary_assembly.annotation.gtf")
names(mcols(gene_annotation))

gene_annotation_df <- as.data.frame(gene_annotation)

# Extract only exon entries
exons <- gene_annotation[gene_annotation$type == "exon"]
names(exons) <- exons$gene_name

# Gene bodies (type == "gene")
gene_bodies <- gene_annotation[gene_annotation$type == "gene"]
names(gene_bodies) <- gene_bodies$gene_name
```

1. 'source' 2. 'type' 3. 'score' 4. 'phase' 5. 'gene_id' 6. 'gene_type' 7. 'gene_status' 8. 'gene_name' 9. 'level' 10. 'havana_gene' 11. 'transcript_id' 12. 'transcript_type' 13. 'transcript_status' 14. 'transcript_name' 15. 'transcript_support_level' 16. 'tag' 17. 'havana_transcript' 18. 'exon_number' 19. 'exon_id' 20. 'protein_id' 21. 'ccdsid' 22. 'ont'

2.1 Detected TEs overlapping exons

[10]:
```
# ================================================================================
# Genomic Classification of Transposable Elements (TEs)
# ================================================================================
# This script iterates through samples and tools (including the simulated data)
# to map TE loci against exons. It identifies overlapping transcripts and
# classifies each TE as exonic, intronic, or intergenic.

# Initialize nested lists to store results per sample and tool
overlapDfList_TE      <- list()
classificationList_TE <- list()

for (sample in samples) {
    overlapDfList_TE[[sample]]      <- list()
    classificationList_TE[[sample]] <- list()

    # Include "simulated" data as a pseudo-tool alongside actual tools
    tools_and_sim <- c(names(objList), "simulated")

    for (tool in tools_and_sim) {
        print(paste("Processing:", sample, "-", tool))

        ### 1. Extract TE Loci & Build GRanges ---------------------------------
        # Fetch the correct object depending on whether it's simulated or real␣
 ↪data
        if (tool == "simulated") {
            obj <- splatter_objs[[sample]]
        } else {
            obj <- objList[[tool]][[sample]]
        }

        # Filter genomic coordinates from the conversion table for detected␣
 ↪features
        bed_TE_tool <- conversion_table[
            conversion_table$stellarscopeID %in% Features(obj),
            c("chr", "start", "end", "strand", "stellarscopeID")
        ]
        rownames(bed_TE_tool) <- bed_TE_tool$stellarscopeID

        # Convert dataframe to a GRanges object for genomic overlap analysis
        gr_TE_tool <- makeGRangesFromDataFrame(bed_TE_tool)
        names(gr_TE_tool) <- rownames(bed_TE_tool)


        ### 2. Calculate Exon Overlaps -----------------------------------------
        # Find overlaps between TE features and exon annotations
```

```
        exonOverlaps <- findOverlaps(gr_TE_tool, exons, minoverlap = 1)

        # Compute exact overlap length in base pairs
        overlap_bp <- width(pintersect(
            gr_TE_tool[queryHits(exonOverlaps)],
            exons[subjectHits(exonOverlaps)]
        ))

        # Build overlap dataframe and flag features present in the simulation
        overlap_df <- data.frame(
            TE            = names(gr_TE_tool)[queryHits(exonOverlaps)],
            exon          = names(exons)[subjectHits(exonOverlaps)],
            gene_id       = mcols(exons)$gene_id[subjectHits(exonOverlaps)],
            transcript_id =␣
↪mcols(exons)$transcript_id[subjectHits(exonOverlaps)],
            gene_name     = mcols(exons)$gene_name[subjectHits(exonOverlaps)],
            overlap_bp    = overlap_bp
        ) %>%
            dplyr::mutate(
                TEinSimulation   = TE %in% Features(splatter_objs[[sample]]),
                geneInSimulation = gene_name %in%␣
↪Features(splatter_objs_genes[[sample]])
            ) %>%
            # Resolve multi-mapping TEs by keeping only the highest-overlapping␣
↪exon
            dplyr::group_by(TE) %>%
            dplyr::slice_max(overlap_bp, n = 1, with_ties = FALSE) %>%
            dplyr::ungroup()


        ### 3. Hierarchical Genomic Classification ----------------------------
        # Exonic: Overlaps at least 1bp of an exon
        exonic_TEs <- unique(overlap_df$TE)

        # Genic: Overlaps any part of the broader gene body
        genic_TEs <- unique(names(gr_TE_tool)[queryHits(
            findOverlaps(gr_TE_tool, gene_bodies, minoverlap = 1)
        )])

        # Intronic: Within gene body but completely misses exons
        intronic_TEs   <- setdiff(genic_TEs, exonic_TEs)
        all_TEs        <- names(gr_TE_tool)

        # Intergenic: Falls entirely outside known gene bodies
        intergenic_TEs <- setdiff(all_TEs, genic_TEs)
```

```
        ### 4. Construct Final Classification Table ----------------------------
        classification_df <- data.frame(
            TE       = all_TEs,
            category = dplyr::case_when(
                all_TEs %in% exonic_TEs   ~ "exonic",
                all_TEs %in% intronic_TEs ~ "intronic",
                TRUE                      ~ "intergenic"
            )
        ) %>%
            # Append exon-specific metadata to TEs classified as exonic␣
  ↪(overlap length, whether the gene is simulated)
            dplyr::left_join(
                dplyr::select(overlap_df, TE, gene_id, gene_name,
                              transcript_id, overlap_bp, geneInSimulation),
                by = "TE"
            ) %>%
            dplyr::mutate(
                TEinSimulation = TE %in% Features(splatter_objs[[sample]])
            )


        ### 5. Store Output␣
  ↪-----------------------------------------------------
        overlapDfList_TE[[sample]][[tool]]      <- overlap_df
        classificationList_TE[[sample]][[tool]] <- classification_df
    }
}
```

```
[1] "Processing: all - STARsolo_TE"
```

```
Warning message in .merge_two_Seqinfo_objects(x, y):
"Each of the 2 combined objects has sequence levels not in the other:
  - in 'x': GL456382.1
  - in 'y': chrM, GL456216.1, GL456219.1, GL456221.1, GL456239.1, GL456350.1,
GL456381.1, JH584292.1, JH584293.1, JH584294.1, JH584295.1, JH584296.1,
JH584298.1, JH584299.1, JH584303.1
  Make sure to always combine/compare objects based on the same reference
  genome (use suppressWarnings() to suppress this warning)."
Warning message in .merge_two_Seqinfo_objects(x, y):
"Each of the 2 combined objects has sequence levels not in the other:
  - in 'x': GL456382.1
  - in 'y': chrM, GL456216.1, GL456219.1, GL456221.1, GL456239.1, GL456350.1,
GL456381.1, JH584292.1, JH584293.1, JH584294.1, JH584295.1, JH584296.1,
JH584298.1, JH584299.1, JH584303.1
  Make sure to always combine/compare objects based on the same reference
  genome (use suppressWarnings() to suppress this warning)."
Warning message in .merge_two_Seqinfo_objects(x, y):
"Each of the 2 combined objects has sequence levels not in the other:
```

```
  - in 'x': GL456382.1
  - in 'y': chrM, GL456216.1, GL456219.1, GL456221.1, GL456239.1, GL456350.1,
GL456381.1, JH584292.1, JH584293.1, JH584294.1, JH584295.1, JH584296.1,
JH584298.1, JH584299.1, JH584303.1
  Make sure to always combine/compare objects based on the same reference
  genome (use suppressWarnings() to suppress this warning)."

[1] "Processing: all - STARsolo_TE_EM"

Warning message in .merge_two_Seqinfo_objects(x, y):
"Each of the 2 combined objects has sequence levels not in the other:
  - in 'x': GL456382.1, JH584300.1, JH584301.1
  - in 'y': chrM, GL456216.1, GL456239.1, GL456381.1, JH584292.1, JH584294.1,
JH584295.1
  Make sure to always combine/compare objects based on the same reference
  genome (use suppressWarnings() to suppress this warning)."
Warning message in .merge_two_Seqinfo_objects(x, y):
"Each of the 2 combined objects has sequence levels not in the other:
  - in 'x': GL456382.1, JH584300.1, JH584301.1
  - in 'y': chrM, GL456216.1, GL456239.1, GL456381.1, JH584292.1, JH584294.1,
JH584295.1
  Make sure to always combine/compare objects based on the same reference
  genome (use suppressWarnings() to suppress this warning)."
Warning message in .merge_two_Seqinfo_objects(x, y):
"Each of the 2 combined objects has sequence levels not in the other:
  - in 'x': GL456382.1, JH584300.1, JH584301.1
  - in 'y': chrM, GL456216.1, GL456239.1, GL456381.1, JH584292.1, JH584294.1,
JH584295.1
  Make sure to always combine/compare objects based on the same reference
  genome (use suppressWarnings() to suppress this warning)."

[1] "Processing: all - SoloTE_unique"
[1] "Processing: all - SoloTE_thr2"
[1] "Processing: all - SoloTE_thr1"
[1] "Processing: all - SoloTE_thr0"
[1] "Processing: all - Stellarscope"

Warning message in .merge_two_Seqinfo_objects(x, y):
"Each of the 2 combined objects has sequence levels not in the other:
  - in 'x': GL456382.1
  - in 'y': chrM, GL456216.1, GL456219.1, GL456239.1, GL456350.1, GL456381.1,
JH584292.1, JH584293.1, JH584294.1, JH584295.1, JH584296.1, JH584298.1,
JH584299.1, JH584303.1
  Make sure to always combine/compare objects based on the same reference
  genome (use suppressWarnings() to suppress this warning)."
Warning message in .merge_two_Seqinfo_objects(x, y):
"Each of the 2 combined objects has sequence levels not in the other:
  - in 'x': GL456382.1
  - in 'y': chrM, GL456216.1, GL456219.1, GL456239.1, GL456350.1, GL456381.1,
JH584292.1, JH584293.1, JH584294.1, JH584295.1, JH584296.1, JH584298.1,
```

```
JH584299.1, JH584303.1
  Make sure to always combine/compare objects based on the same reference
  genome (use suppressWarnings() to suppress this warning)."
Warning message in .merge_two_Seqinfo_objects(x, y):
"Each of the 2 combined objects has sequence levels not in the other:
  - in 'x': GL456382.1
  - in 'y': chrM, GL456216.1, GL456219.1, GL456239.1, GL456350.1, GL456381.1,
JH584292.1, JH584293.1, JH584294.1, JH584295.1, JH584296.1, JH584298.1,
JH584299.1, JH584303.1
  Make sure to always combine/compare objects based on the same reference
  genome (use suppressWarnings() to suppress this warning)."
[1] "Processing: all - simulated"
```

[12]:
```
head(overlapDfList_TE[[sample]][[tool]])
head(classificationList_TE[[sample]][[tool]])
```

A tibble: 6 × 8

TE <chr>	exon <chr>	gene_id <chr>	transcript_id <chr>
AmnSINE1-dup1253	Malt1	ENSMUSG00000032688.7	ENSMUST00000049248.5
AmnSINE1-dup163	Gm25966	ENSMUSG00000087794.1	ENSMUST00000157169.1
AmnSINE1-dup389	Gm26486	ENSMUSG00000088646.1	ENSMUST00000158021.1
AmnSINE1-dup576	Mob2	ENSMUSG00000025147.7	ENSMUST00000211633.1
Arthur1-dup4	4921511E07Rik	ENSMUSG00000104133.1	ENSMUST00000194790.1
B1-Mus1-dup44044	Gm17545	ENSMUSG00000091159.1	ENSMUST00000168332.1

A data.frame: 6 × 8

	TE <chr>	category <chr>	gene_id <chr>	gene_name <chr>	transcript_id <chr>	overlap_bp <int>	gen <lg
1	X3-LINE	intronic	NA	NA	NA	NA	NA
2	IAPEz-int-dup9	intergenic	NA	NA	NA	NA	NA
3	RLTR41-dup1	intergenic	NA	NA	NA	NA	NA
4	MT2-Mm-dup5	intergenic	NA	NA	NA	NA	NA
5	Looper-dup1	intergenic	NA	NA	NA	NA	NA
6	B1-Mus1-dup96	intergenic	NA	NA	NA	NA	NA

3 Checkpoint

[1]:
```
#save.image("workspaces/mm_RA_wGenes_afterClassification.Rdata")
load("workspaces/mm_RA_wGenes_afterClassification.Rdata")
```

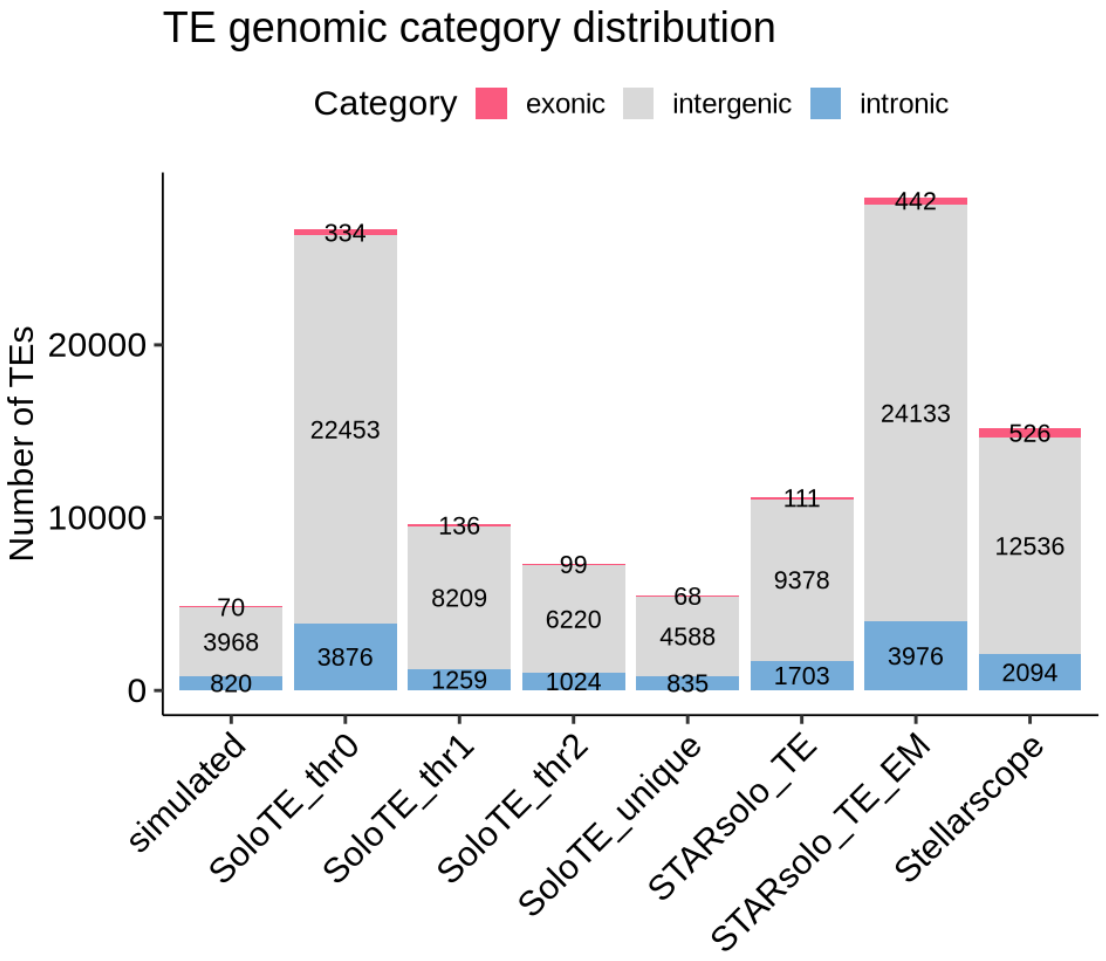
[2]:
```
gc()
```

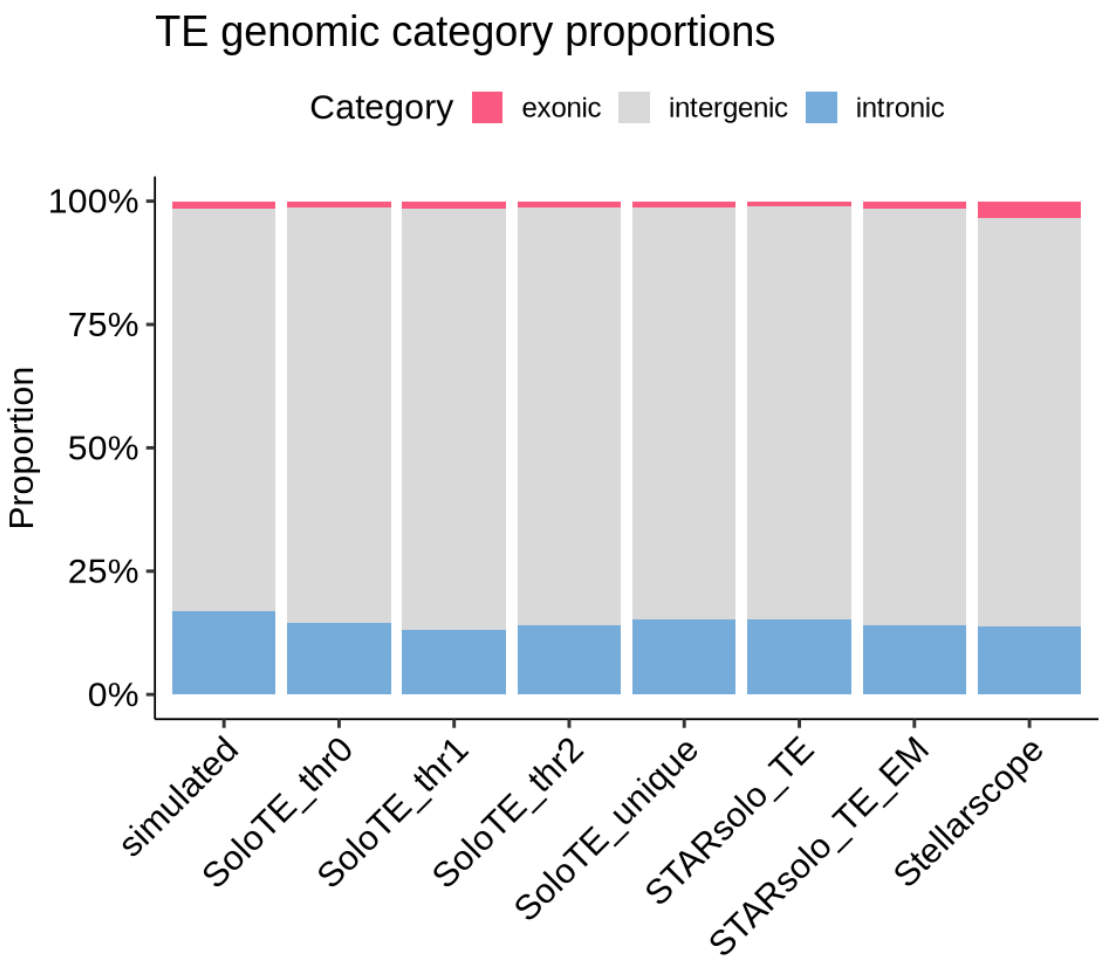
A matrix: 2 × 6 of type dbl

	used	(Mb)	gc trigger	(Mb)	max used	(Mb)
Ncells	21000952	1121.6	39486914	2108.9	21327093	1139.0
Vcells	485192706	3701.8	662749398	5056.4	485367112	3703.1

[8]:
```
#   Flatten classificationList_TE into one df
classification_all <- purrr::imap_dfr(classificationList_TE,␣
 ↪function(sample_list, sample){
```

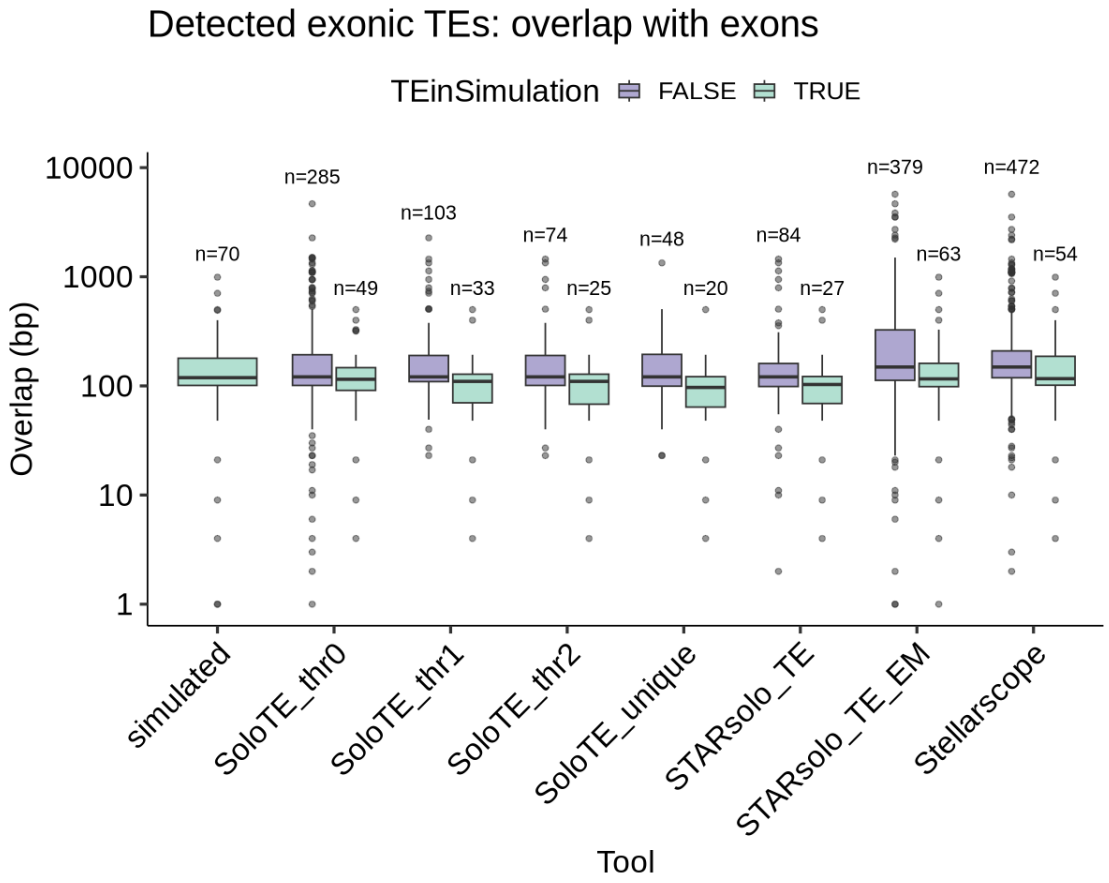
```
    purrr::imap_dfr(sample_list, function(df, tool){
        df %>% dplyr::mutate(sample = sample, tool = tool)
    })
})

options(repr.plot.width=8, repr.plot.height=7)

# Define a base text size for all plots
text_size <- 18

# Category counts per tool/sample (stacked bar)
classification_all %>%
    dplyr::count(sample, tool, category) %>%
    ggplot(aes(x = tool, y = n, fill = category)) +
    geom_col(position = "stack") +
    geom_text(aes(label = n), position = position_stack(vjust = 0.5), size = 4.
 ↪5) + # Larger text
    scale_fill_manual(values = c(exonic = "#fa5a7f", intronic = "#75acd9",␣
 ↪intergenic = "grey85")) +
    labs(title = "TE genomic category distribution",
         x = NULL, y = "Number of TEs", fill = "Category") +
    theme_pubr(base_size = text_size) +
    theme(axis.text.x = element_text(angle = 45, hjust = 1))
ggsave(paste0("figures/figures_", dataset_id, "_", sample,"/
 ↪TEgenomicCategory_numbers.pdf"), device="pdf", width=8, height=7)

# Category proportions per tool/sample (filled bar)
classification_all %>%
    dplyr::count(sample, tool, category) %>%
    ggplot(aes(x = tool, y = n, fill = category)) +
    geom_col(position = "fill") +
    scale_y_continuous(labels = scales::percent) +
    scale_fill_manual(values = c(exonic = "#fa5a7f", intronic = "#75acd9",␣
 ↪intergenic = "grey85")) +
    labs(title = "TE genomic category proportions",
         x = NULL, y = "Proportion", fill = "Category") +
    theme_pubr(base_size = text_size) +
    theme(axis.text.x = element_text(angle = 45, hjust = 1))
ggsave(paste0("figures/figures_", dataset_id, "_", sample,"/
 ↪TEgenomicCategory_percentages.pdf"), device="pdf", width=8, height=7)
```


TE genomic category distribution
Category
exonic
intergenic
intronic
Number of TEs
20000
10000
0
simulated
SoloTE_thr0
SoloTE_thr1
SoloTE_thr2
SoloTE_unique
STARsolo_TE
STARsolo_TE_EM
Stellarscope
70
3968
820
334
22453
3876
136
8209
1259
99
6220
1024
68
4588
835
111
9378
1703
442
24133
3976
526
12536
2094



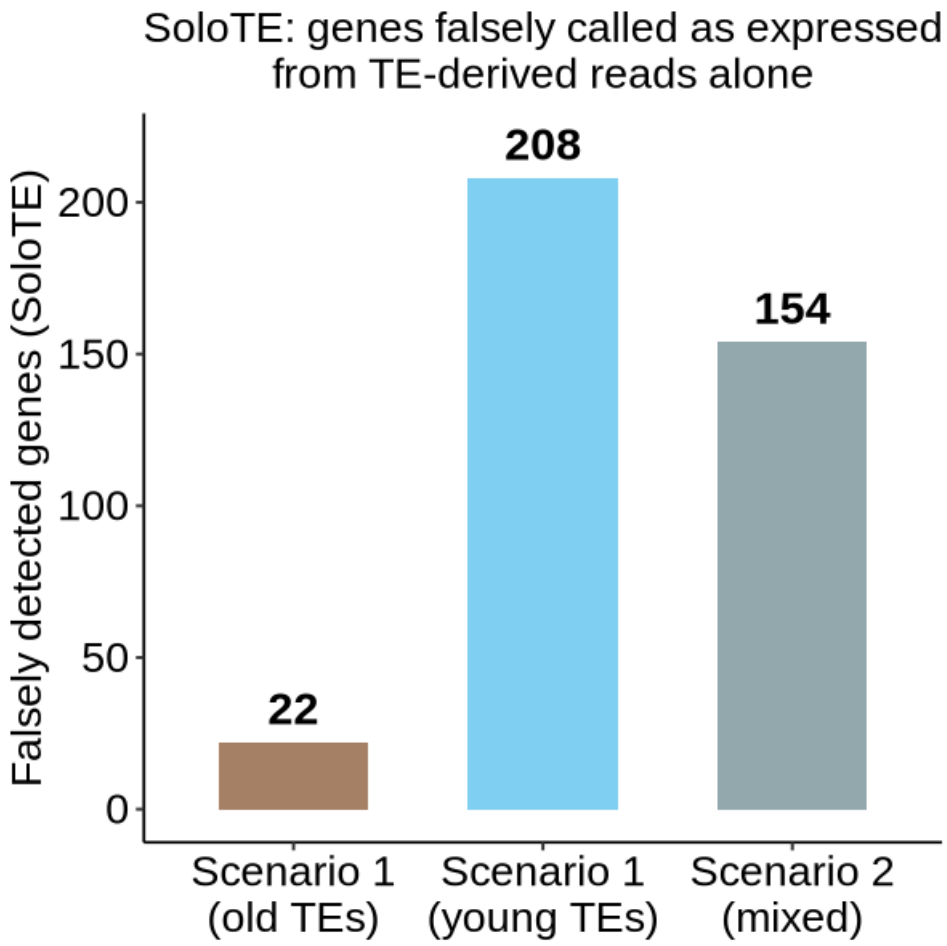
[11]:
```
options(repr.plot.width=10, repr.plot.height=8)
text_size <- 20

classification_all %>%
    dplyr::filter(category == "exonic") %>%
    ggplot(aes(x = tool, y = overlap_bp, fill = TEinSimulation)) +
    geom_boxplot(alpha = 0.5) +
    stat_summary(
        fun.data = function(y) {
            data.frame(
                y = max(y, na.rm = TRUE) * 1.05,  # slightly above max
                label = paste0("n=", length(y))
            )
        },
        geom = "text",
        size = 4.5,
```

```
        position = position_dodge(0.75),
        vjust = 0
    ) +
    scale_y_continuous(transform = "log10") +
    scale_fill_manual(values = c("TRUE" = "#67c1a3", "FALSE" = "#5e50a1")) +
    labs(
        title = "Detected exonic TEs: overlap with exons",
        x = "Tool",
        y = "Overlap (bp)"
    ) +
    theme_pubr(base_size = text_size) +
    theme(axis.text.x = element_text(angle = 45, hjust = 1))
ggsave(paste0("figures/figures_", dataset_id, "_", sample,"/
 ↪TEgene_overlap_FPandTP.pdf"), device="pdf", width=10, height=8)
```



[20]:
```
# plot number of genes detected by SoloTE in scenario 1 and 2
options(repr.plot.width=5, repr.plot.height=5)

detectedGenesSoloTE <- list("Scenario1-old" = 22,
```

```
                              "Scenario1-young" = 208,
                              "Scenario2-mixed" = 154)

df <- data.frame(
  scenario = factor(names(detectedGenesSoloTE),
                    levels = names(detectedGenesSoloTE),
                    labels = c("Scenario 1\n(old TEs)",
                               "Scenario 1\n(young TEs)",
                               "Scenario 2\n(mixed)")),
  n_genes = unlist(detectedGenesSoloTE)
)

ggplot(df, aes(x = scenario, y = n_genes, fill = scenario)) +
  geom_col(width = 0.6) +
  geom_text(aes(label = n_genes), vjust = -0.6, fontface = "bold", size = 6) +
  scale_fill_manual(values = c("#A58065", "#7FCFF2", "#92A8AC")) +
  labs(
    title = "SoloTE: genes falsely called as expressed\nfrom TE-derived reads␣
 ↪alone",
    x = NULL,
    y = "Falsely detected genes (SoloTE)"
  ) +
  ylim(0, max(df$n_genes) * 1.05) +
  theme_pubr() +
  theme(
    legend.position = "none",
    plot.title = element_text(hjust = 0.5, size = 16),
    text = element_text(size = 16)
  )

ggsave(paste0("figures/figures_", dataset_id, "_", sample,"/
 ↪SoloTE_incorrectlyDetectedGenes_scenario1and2.pdf"), device="pdf", width=5,␣
 ↪height=5)
```



4 Gene intersection plots

[64]:
```r
# compute intersection between list of loci of each tool and gene annotation
for (sample in samples){

    for(tool in names(objList)){

        print(tool)
        overlapDfList_TE[[sample]][[tool]]$detection <- NA


␣
↪overlapDfList_TE[[sample]][[tool]]$detection[overlapDfList_TE[[sample]][[tool]]$TE␣
↪%in% TPlist[[tool]]] <- "TPs"

␣
↪overlapDfList_TE[[sample]][[tool]]$detection[overlapDfList_TE[[sample]][[tool]]$TE␣
↪%in% FPlist[[tool]]] <- "FPs"
```

```
    }
}
```

```
[1] "STARsolo_TE"
[1] "STARsolo_TE_EM"
[1] "SoloTE_unique"
[1] "SoloTE_thr2"
[1] "SoloTE_thr1"
[1] "SoloTE_thr0"
[1] "Stellarscope"
```

[ ]:
```
percPosOverlapGenesDf <- NULL

for (sample in samples){

    for(tool in names(objList)){

        print(tool)

        df <- cbind(melt(table(overlapDfList_TE[[sample]][[tool]]$detection,␣
 ↪overlapDfList_TE[[sample]][[tool]]$geneInSimulation )), tool)
        colnames(df) <- c("Detection", "ExpressedGene", "Number", "Tool")
        print(df)
        percPosOverlapGenesDf <- rbind(percPosOverlapGenesDf, df)

    }
}
```

```
[1] "STARsolo_TE"
  Detection ExpressedGene Number        Tool
1       FPs         FALSE     59 STARsolo_TE
2       TPs         FALSE     23 STARsolo_TE
3       FPs          TRUE     25 STARsolo_TE
4       TPs          TRUE      4 STARsolo_TE
[1] "STARsolo_TE_EM"
  Detection ExpressedGene Number           Tool
1       FPs         FALSE    225 STARsolo_TE_EM
2       TPs         FALSE     47 STARsolo_TE_EM
3       FPs          TRUE    154 STARsolo_TE_EM
4       TPs          TRUE     16 STARsolo_TE_EM
[1] "SoloTE_unique"
  Detection ExpressedGene Number          Tool
1       FPs         FALSE     27 SoloTE_unique
2       TPs         FALSE     19 SoloTE_unique
3       FPs          TRUE     21 SoloTE_unique
4       TPs          TRUE      1 SoloTE_unique
[1] "SoloTE_thr2"
```

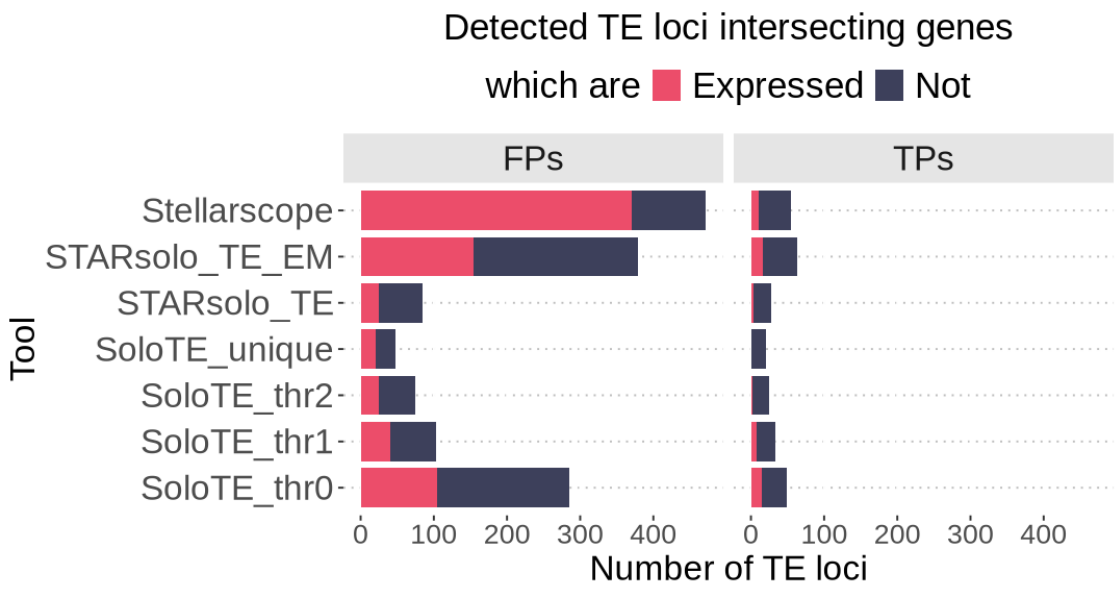
```
  Detection ExpressedGene Number        Tool
1       FPs         FALSE     49 SoloTE_thr2
2       TPs         FALSE     23 SoloTE_thr2
3       FPs          TRUE     25 SoloTE_thr2
4       TPs          TRUE      2 SoloTE_thr2
[1] "SoloTE_thr1"
  Detection ExpressedGene Number        Tool
1       FPs         FALSE     63 SoloTE_thr1
2       TPs         FALSE     25 SoloTE_thr1
3       FPs          TRUE     40 SoloTE_thr1
4       TPs          TRUE      8 SoloTE_thr1
[1] "SoloTE_thr0"
  Detection ExpressedGene Number        Tool
1       FPs         FALSE    181 SoloTE_thr0
2       TPs         FALSE     34 SoloTE_thr0
3       FPs          TRUE    104 SoloTE_thr0
4       TPs          TRUE     15 SoloTE_thr0
[1] "Stellarscope"
  Detection ExpressedGene Number         Tool
1       FPs         FALSE    102 Stellarscope
2       TPs         FALSE     44 Stellarscope
3       FPs          TRUE    370 Stellarscope
4       TPs          TRUE     10 Stellarscope
```

[66]:
```
percPosOverlapGenesDf$ExpressedGene <- as.
  ↪character(percPosOverlapGenesDf$ExpressedGene)
```

[ ]:
```
options(repr.plot.width=9, repr.plot.height=4.75)

ggplot(percPosOverlapGenesDf, aes(y=Tool, x=Number, fill=ExpressedGene)) +
    facet_wrap(~Detection) +
    geom_col(width = 0.85) +
    scale_fill_manual(values=c("TRUE"="#EC4D6A","FALSE"="#3d405b"),
                      labels = c("TRUE"="Expressed","FALSE"="Not"),
                      breaks=c("TRUE", "FALSE"), name="which are") +
    theme_pubclean() +
    ggtitle("Detected TE loci intersecting genes") +
    xlab("Number of TE loci") +
  theme(text=element_text(size=20),
        axis.text.y = element_text(size=20),
        axis.label = element_text(size=18),
        legend.title = element_text(size=21),
        legend.text = element_text(size=21),
        strip.text = element_text(size = 20),
        strip.background = element_rect(fill="grey90"),
        plot.title = element_text(size=21, hjust=0.5),
        plot.subtitle = element_text(size=20, hjust=0.5))
```

```
ggsave(paste0("figures/figures_", dataset_id, "_", sample,"/
 ↪detectedTEsIntersectingGenes.pdf"),
       device="pdf", width=9, height=4.5)
```

```
Warning message in plot_theme(plot):
"The `axis.label` theme element is not defined in the element
hierarchy."
```



4.1 Detected Genes overlapping TEs

[68]:
```
gc()

bed_TE <- conversion_table[,c("chr","start","end","strand","stellarscopeID")]
#rownames(bed_TE) <- bed_TE$stellarscopeID

gr_TE <- makeGRangesFromDataFrame(bed_TE, keep.extra.columns = TRUE )
```

A matrix: 2 × 6 of type dbl

	used	(Mb)	gc trigger	(Mb)	max used	(Mb)
Ncells	31429163	1678.5	55645654	2971.8	55645654	2971.8
Vcells	498574866	3803.9	877976232	6698.5	841749691	6422.1

[ ]:
```
overlapDfList_GENE <- list()
overlapDfList_GENE_unique <- list()

# compute intersection between list of loci of each tool and gene annotation
for (sample in samples){
    overlapDfList_GENE[[sample]] <- list()
```

```
    for(tool in names(objGeneList)){

        print(tool)
        obj <- objGeneList[[tool]][[sample]]

        # get detected TE gr
        gr_GENE_tool <- exons[exons$gene_name %in% Features(obj)]

        exonOverlaps <- findOverlaps(gr_GENE_tool, gr_TE, minoverlap = 1)

        queryHits <- queryHits(exonOverlaps)
        subjectHits <- subjectHits(exonOverlaps)

        overlap_df <- data.frame(
            Exon = names(gr_GENE_tool)[queryHits],
            gene_id = mcols(gr_GENE_tool)$gene_id[queryHits],
            gene_name = mcols(gr_GENE_tool)$gene_name[queryHits],
            TE = mcols(gr_TE)$stellarscopeID[subjectHits]
        )

        # column indicating wether the overlapping TE is in the simulated TE␣
↪matrix
        overlap_df$TEInSimulation <- overlap_df$TE %in%␣
↪Features(splatter_objs[[sample]])

        overlapDfList_GENE[[sample]][[tool]] <- overlap_df

        # create df with only the unique lines, prioritizing TEs that are in␣
↪the simulation
        overlapDfList_GENE_unique[[sample]][[tool]]  <-␣
↪overlapDfList_GENE[[sample]][[tool]] %>% distinct()
        overlapDfList_GENE_unique[[sample]][[tool]]  <-␣
↪overlapDfList_GENE_unique[[sample]][[tool]]  %>%
                group_by(gene_name) %>%
                slice_max(order_by = TEInSimulation, n = 1, with_ties = FALSE)␣
↪%>%
                ungroup()

    }
}
```

```
[1] "STARsolo_TE"
```

```
Warning message in .merge_two_Seqinfo_objects(x, y):
"Each of the 2 combined objects has sequence levels not in the other:
  - in 'x': chrM, JH584295.1
  - in 'y': GL456213.1, GL456359.1, GL456360.1, GL456366.1, GL456367.1,
GL456368.1, GL456370.1, GL456378.1, GL456379.1, GL456382.1, GL456383.1,
```

```
GL456387.1, GL456389.1, GL456390.1, GL456392.1, GL456393.1, GL456394.1,
GL456396.1, JH584300.1, JH584301.1, JH584302.1
  Make sure to always combine/compare objects based on the same reference
  genome (use suppressWarnings() to suppress this warning)."

[1] "STARsolo_TE_EM"

Warning message in .merge_two_Seqinfo_objects(x, y):
"Each of the 2 combined objects has sequence levels not in the other:
  - in 'x': chrM, JH584295.1
  - in 'y': GL456213.1, GL456359.1, GL456360.1, GL456366.1, GL456367.1,
GL456368.1, GL456370.1, GL456378.1, GL456379.1, GL456382.1, GL456383.1,
GL456387.1, GL456389.1, GL456390.1, GL456392.1, GL456393.1, GL456394.1,
GL456396.1, JH584300.1, JH584301.1, JH584302.1
  Make sure to always combine/compare objects based on the same reference
  genome (use suppressWarnings() to suppress this warning)."

[1] "SoloTE_unique"

Warning message in .merge_two_Seqinfo_objects(x, y):
"Each of the 2 combined objects has sequence levels not in the other:
  - in 'x': chrM, JH584295.1
  - in 'y': GL456213.1, GL456359.1, GL456360.1, GL456366.1, GL456367.1,
GL456368.1, GL456370.1, GL456378.1, GL456379.1, GL456382.1, GL456383.1,
GL456387.1, GL456389.1, GL456390.1, GL456392.1, GL456393.1, GL456394.1,
GL456396.1, JH584300.1, JH584301.1, JH584302.1
  Make sure to always combine/compare objects based on the same reference
  genome (use suppressWarnings() to suppress this warning)."

[1] "SoloTE_thr2"

Warning message in .merge_two_Seqinfo_objects(x, y):
"Each of the 2 combined objects has sequence levels not in the other:
  - in 'x': chrM, JH584295.1
  - in 'y': GL456213.1, GL456359.1, GL456360.1, GL456366.1, GL456367.1,
GL456368.1, GL456370.1, GL456378.1, GL456379.1, GL456382.1, GL456383.1,
GL456387.1, GL456389.1, GL456390.1, GL456392.1, GL456393.1, GL456394.1,
GL456396.1, JH584300.1, JH584301.1, JH584302.1
  Make sure to always combine/compare objects based on the same reference
  genome (use suppressWarnings() to suppress this warning)."

[1] "SoloTE_thr1"

Warning message in .merge_two_Seqinfo_objects(x, y):
"Each of the 2 combined objects has sequence levels not in the other:
  - in 'x': chrM, JH584295.1
  - in 'y': GL456213.1, GL456359.1, GL456360.1, GL456366.1, GL456367.1,
GL456368.1, GL456370.1, GL456378.1, GL456379.1, GL456382.1, GL456383.1,
GL456387.1, GL456389.1, GL456390.1, GL456392.1, GL456393.1, GL456394.1,
GL456396.1, JH584300.1, JH584301.1, JH584302.1
  Make sure to always combine/compare objects based on the same reference
  genome (use suppressWarnings() to suppress this warning)."
```

```
[1] "SoloTE_thr0"
```

```
Warning message in .merge_two_Seqinfo_objects(x, y):
"Each of the 2 combined objects has sequence levels not in the other:
  - in 'x': chrM, JH584295.1
  - in 'y': GL456213.1, GL456359.1, GL456360.1, GL456366.1, GL456367.1,
GL456368.1, GL456370.1, GL456378.1, GL456379.1, GL456382.1, GL456383.1,
GL456387.1, GL456389.1, GL456390.1, GL456392.1, GL456393.1, GL456394.1,
GL456396.1, JH584300.1, JH584301.1, JH584302.1
  Make sure to always combine/compare objects based on the same reference
  genome (use suppressWarnings() to suppress this warning)."
```

[70]:
```
## Detected genes overlapping TE loci
# compute intersection between list of loci of each tool and gene annotation
for (sample in samples){

    for(tool in names(objGeneList)){

        print(tool)
        overlapDfList_GENE_unique[[sample]][[tool]]$detection <- NA


      ␣
 ↪overlapDfList_GENE_unique[[sample]][[tool]]$detection[overlapDfList_GENE_unique[[sample]][[
 ↪%in% TPgenelist[[tool]]] <- "TPs"
      ␣
 ↪overlapDfList_GENE_unique[[sample]][[tool]]$detection[overlapDfList_GENE_unique[[sample]][[
 ↪%in% FPgenelist[[tool]]] <- "FPs"

    }
}
```

```
[1] "STARsolo_TE"
[1] "STARsolo_TE_EM"
[1] "SoloTE_unique"
[1] "SoloTE_thr2"
[1] "SoloTE_thr1"
[1] "SoloTE_thr0"
```

[7]:
```
head(overlapDfList_GENE_unique[[sample]][[tool]])
```

A tibble: 6 × 6

Exon <chr>	gene_id <chr>	gene_name <chr>	TE <chr>	TEInSim <lgl>
0610031O16Rik	ENSMUSG00000099146.7	0610031O16Rik	B4-dup13192	FALSE
0610040J01Rik	ENSMUSG00000060512.7	0610040J01Rik	MT2B-dup3804	FALSE
1190002N15Rik	ENSMUSG00000045414.7	1190002N15Rik	L3b-dup1148	FALSE
1600029O15Rik	ENSMUSG00000057818.8	1600029O15Rik	RCHARR1-dup3433	FALSE
1700003I16Rik	ENSMUSG00000107526.1	1700003I16Rik	MERV1-I-int-dup438	FALSE
1700016L21Rik	ENSMUSG00000101483.1	1700016L21Rik	MLT1J2-dup42	FALSE

[ ]:
```
percPosOverlapTEsDf <- NULL

for (sample in samples){

    for(tool in names(objGeneList)){

        print(tool)

        df <-␣
 ↪cbind(melt(table(overlapDfList_GENE_unique[[sample]][[tool]]$detection,␣
 ↪overlapDfList_GENE_unique[[sample]][[tool]]$TEInSimulation )), tool)
        colnames(df) <- c("Detection", "ExpressedTE", "Number", "Tool")
        print(df)
        percPosOverlapTEsDf <- rbind(percPosOverlapTEsDf, df)

    }
}
```

```
[1] "STARsolo_TE"
  Detection ExpressedTE Number        Tool
1       FPs       FALSE      8 STARsolo_TE
2       TPs       FALSE    879 STARsolo_TE
3       FPs        TRUE      0 STARsolo_TE
4       TPs        TRUE     14 STARsolo_TE
[1] "STARsolo_TE_EM"
  Detection ExpressedTE Number           Tool
1       FPs       FALSE    174 STARsolo_TE_EM
2       TPs       FALSE    993 STARsolo_TE_EM
3       FPs        TRUE     21 STARsolo_TE_EM
4       TPs        TRUE     15 STARsolo_TE_EM
[1] "SoloTE_unique"
  Detection ExpressedTE Number          Tool
1       FPs       FALSE    113 SoloTE_unique
2       TPs       FALSE    953 SoloTE_unique
3       FPs        TRUE     22 SoloTE_unique
4       TPs        TRUE     18 SoloTE_unique
[1] "SoloTE_thr2"
  Detection ExpressedTE Number        Tool
1       FPs       FALSE    113 SoloTE_thr2
2       TPs       FALSE    953 SoloTE_thr2
3       FPs        TRUE     22 SoloTE_thr2
4       TPs        TRUE     18 SoloTE_thr2
[1] "SoloTE_thr1"
  Detection ExpressedTE Number        Tool
1       FPs       FALSE    113 SoloTE_thr1
2       TPs       FALSE    953 SoloTE_thr1
3       FPs        TRUE     22 SoloTE_thr1
4       TPs        TRUE     18 SoloTE_thr1
```

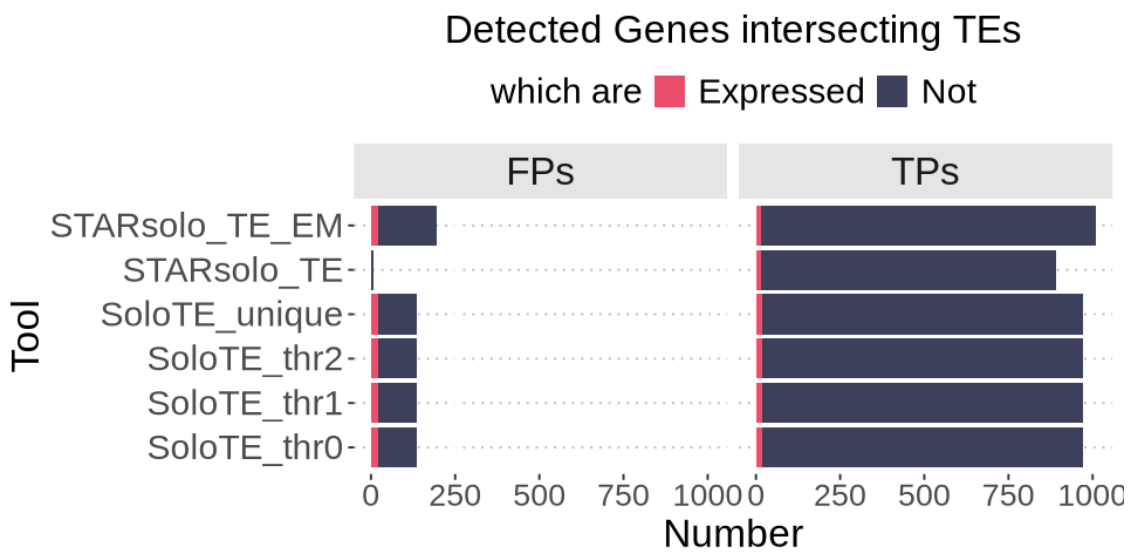
```
[1] "SoloTE_thr0"
  Detection ExpressedTE Number        Tool
1       FPs       FALSE    113 SoloTE_thr0
2       TPs       FALSE    953 SoloTE_thr0
3       FPs        TRUE     22 SoloTE_thr0
4       TPs        TRUE     18 SoloTE_thr0
```

[72]:
```r
percPosOverlapTEsDf$ExpressedTE <- as.character(percPosOverlapTEsDf$ExpressedTE)

options(repr.plot.width=8, repr.plot.height=4)

ggplot(percPosOverlapTEsDf, aes(y=Tool, x=Number, fill=ExpressedTE)) +
    facet_wrap(~Detection) +
    geom_col() +
    scale_fill_manual(values=c("TRUE"="#EC4D6A","FALSE"="#3d405b"),
                      labels = c("TRUE"="Expressed","FALSE"="Not"),
                      breaks=c("TRUE", "FALSE"), name="which are") +
    theme_pubclean() +
    ggtitle("Detected Genes intersecting TEs") +
  theme(text=element_text(size=20),
        axis.text.y = element_text(size=18),
        axis.label = element_text(size=18),
        legend.title = element_text(size=18),
        legend.text = element_text(size=18),
        strip.text = element_text(size = 20),
        strip.background = element_rect(fill="grey90"),
        plot.title = element_text(size=20, hjust=0.5))
# ggsave(paste0("figures/figures_", dataset_id, "_", sample,"/
 ↪detectedGenesIntersectingTEs.pdf"), device="pdf", width=8, height=5)
```

```
Warning message in plot_theme(plot):
"The `axis.label` theme element is not defined in the element
hierarchy."
```

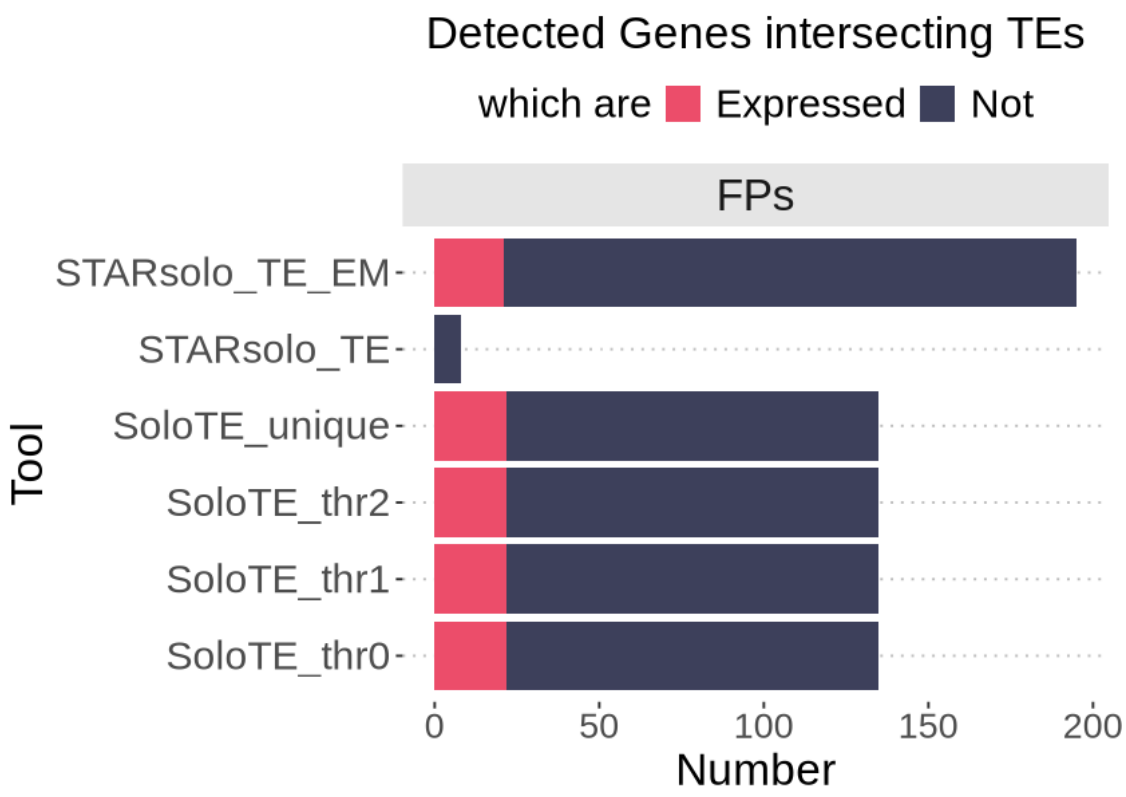


[73]:
```
percPosOverlapTEsDf$ExpressedTE <- as.character(percPosOverlapTEsDf$ExpressedTE)

options(repr.plot.width=7, repr.plot.height=5)

ggplot(percPosOverlapTEsDf[percPosOverlapTEsDf$Detection=="FPs",], aes(y=Tool,␣
 ↪x=Number, fill=ExpressedTE)) +
    facet_wrap(~Detection) +
    geom_col() +
    scale_fill_manual(values=c("TRUE"="#EC4D6A","FALSE"="#3d405b"),
                      labels = c("TRUE"="Expressed","FALSE"="Not"),
                      breaks=c("TRUE", "FALSE"), name="which are") +
    theme_pubclean() +
    ggtitle("Detected Genes intersecting TEs") +
  theme(text=element_text(size=20),
        axis.text.y = element_text(size=18),
        axis.label = element_text(size=18),
        legend.title = element_text(size=18),
        legend.text = element_text(size=18),
        strip.text = element_text(size = 20),
        strip.background = element_rect(fill="grey90"),
        plot.title = element_text(size=20, hjust=0.5))
# ggsave(paste0("figures/figures_", dataset_id, "_", sample,"/
 ↪FPGenesIntersectingTEs.pdf"), device="pdf", width=6, height=5)
```

```
Warning message in plot_theme(plot):
"The `axis.label` theme element is not defined in the element
hierarchy."
```



5 Checkpoint

[74]:
```
#save.image("workspaces/mm_RA_wGenes_evaluation_detection_afterIntersections.
↪Rdata")
```

[2]:
```
load("workspaces/mm_RA_wGenes_evaluation_detection_afterIntersections.Rdata")
```

6 Summary figure

[3]:
```
tool_order <- c(
  "SoloTE unique",
  "STARsolo",
  "STARsolo + EM",
  "Stellarscope"
)
df_gene <- percPosOverlapGenesDf[percPosOverlapGenesDf$Tool %in%
  c("SoloTE_unique","STARsolo_TE","STARsolo_TE_EM","Stellarscope"),]

df_gene_to_TE <- df_gene |>
  mutate(
```

```
    ExpressedGene = as.character(ExpressedGene) == "TRUE",

    Tool = recode(
      Tool,
      "SoloTE_unique"  = "SoloTE unique",
      "STARsolo_TE"    = "STARsolo",
      "STARsolo_TE_EM" = "STARsolo + EM"
    )
  ) |>

  filter(
    Detection == "FPs",
    ExpressedGene
  ) |>

  group_by(Tool) |>
  summarise(
    n_errors = sum(Number),
    .groups = "drop"
  ) |>

  mutate(
    Tool = factor(
      Tool,
      levels = tool_order
    ),

    direction = "Gene \u2192 TE",

    # Treat counts within five loci of the minimum as near-tied
    best_count = min(n_errors, na.rm = TRUE),
    top_tier = n_errors <= best_count + 5
  )

df_gene_to_TE

df_TE <- percPosOverlapTEsDf[percPosOverlapTEsDf$Tool %in%
  c("SoloTE_unique","STARsolo_TE","STARsolo_TE_EM","Stellarscope"),]
df_TE_to_gene <- df_TE |>
  mutate(
    ExpressedTE = as.character(ExpressedTE) == "TRUE",

    Tool = recode(
      Tool,
      "SoloTE_unique"  = "SoloTE unique",
      "SoloTE_thr2"    = "SoloTE max2",
      "SoloTE_thr1"    = "SoloTE max4",
```

```
      "SoloTE_thr0"    = "SoloTE multi",
      "STARsolo_TE"    = "STARsolo",
      "STARsolo_TE_EM" = "STARsolo + EM"
    )
  ) |>

  filter(
    Detection == "FPs",
    ExpressedTE
  ) |>

  group_by(Tool) |>
  summarise(
    n_errors = sum(Number),
    .groups = "drop"
  ) |>

  mutate(
    direction = "TE → gene"
  )
df_TE_to_gene
```

A tibble: 4 × 5

Tool <fct>	n_errors <int>	direction <chr>	best_count <int>	top_tier <lgl>
STARsolo	25	Gene → TE	21	TRUE
STARsolo + EM	154	Gene → TE	21	FALSE
SoloTE unique	21	Gene → TE	21	TRUE
Stellarscope	370	Gene → TE	21	FALSE

A tibble: 3 × 3

Tool <chr>	n_errors <int>	direction <chr>
STARsolo	0	TE → gene
STARsolo + EM	21	TE → gene
SoloTE unique	22	TE → gene

[4]:
```
df_misassignment <- dplyr::bind_rows(
  dplyr::select(
    df_gene_to_TE,
    Tool,
    direction,
    n_errors
  ),

  dplyr::select(
    df_TE_to_gene,
    Tool,
    direction,
    n_errors
```

```
  )
) |>
  dplyr::mutate(
    Tool = factor(
      Tool,
      levels = tool_order
    ),

    direction = factor(
      direction,
      levels = c(
        "Gene → TE",
        "TE → gene"
      )
    )
  ) |>

  dplyr::group_by(direction) |>

  dplyr::mutate(
    best_count = min(
      n_errors,
      na.rm = TRUE
    ),

    top_tier = n_errors <= best_count + 5,

    # Keep zero-error observations visible
    plot_size = pmax(n_errors, 1)
  ) |>

  dplyr::ungroup()
```

[9]:
```
caption_text <- paste(
  paste0(
    "Gene → TE: false-positive TE loci overlapping expressed genes. ",
    "TE → gene: false-positive genes overlapping expressed TEs."
  ),
  paste0(
    "Black outlines indicate methods within five errors ",
    "of the lowest count in each direction."
  ),
  sep = "\n"
)

p_misassignment <- ggplot(
  df_misassignment,
```

```
  aes(
    x = direction,
    y = Tool
  )
) +

  geom_point(
    aes(
      size = plot_size,
      fill = n_errors
    ),
    shape = 21,
    colour = "white",
    stroke = 0.4
  ) +

  geom_point(
    data = filter(
      df_misassignment,
      top_tier
    ),
    aes(size = plot_size),
    shape = 21,
    fill = NA,
    colour = "black",
    stroke = 1.3,
    show.legend = FALSE
  ) +

  geom_text(
    aes(label = n_errors),
    size = 3.5,
    colour = "black"
  ) +

  scale_size_area(
    max_size = 15,
    guide = "none"
  ) +

  scale_fill_gradient(
    low = "#FFEDAB",
    high = "#EE8F90",
    name = "Number of errors"
  ) +

  scale_x_discrete(
```

```
    labels = c(
      "Gene → TE" =
        "Gene-derived reads →\nfalse-positive TE loci",

      "TE → gene" =
        "TE-derived reads →\nfalse-positive genes"
    )
  ) +

  scale_y_discrete(
    limits = rev(tool_order),
    drop = TRUE
  ) +

  labs(
    x = NULL,
    y = NULL,
    caption = caption_text
  ) +

  theme_minimal(base_size = 14) +

  theme(
    panel.grid.minor = element_blank(),

    panel.grid.major = element_line(
      colour = "grey90",
      linewidth = 0.35
    ),

    axis.text.y = element_text(
      colour = "black",
      size = 12
    ),

    axis.text.x = element_text(
      colour = "black",
      face = "bold",
      size = 11,
      lineheight = 1.1,
      margin = margin(t = 8)
    ),

    legend.position = "bottom",

    plot.caption = element_text(
      hjust = 0,
```

```
      colour = "grey30",
      size = 10,
      lineheight = 1.2,
      margin = margin(t = 14)
    ),

    plot.margin = margin(
      t = 10,
      r = 15,
      b = 15,
      l = 10
    )
  )

options(repr.plot.width=7, repr.plot.height=6)
p_misassignment

ggsave("figures/summaryGeneMis.pdf", device = "pdf", width=7, height=6)
```

```
Warning message in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y,
:
"conversion failure on 'Gene-derived reads →' in 'mbcsToSbcs': dot substituted
for <e2>"
Warning message in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y,
:
"conversion failure on 'Gene-derived reads →' in 'mbcsToSbcs': dot substituted
for <86>"
Warning message in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y,
:
"conversion failure on 'Gene-derived reads →' in 'mbcsToSbcs': dot substituted
for <92>"
Warning message in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y,
:
"conversion failure on 'TE-derived reads →' in 'mbcsToSbcs': dot substituted for
<e2>"
Warning message in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y,
:
"conversion failure on 'TE-derived reads →' in 'mbcsToSbcs': dot substituted for
<86>"
Warning message in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y,
:
"conversion failure on 'TE-derived reads →' in 'mbcsToSbcs': dot substituted for
<92>"
Warning message in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y,
:
"conversion failure on 'Gene → TE: false-positive TE loci overlapping expressed
genes. TE → gene: false-positive genes overlapping expressed TEs.' in
'mbcsToSbcs': dot substituted for <e2>"
```

```
Warning message in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y,
:
"conversion failure on 'Gene → TE: false-positive TE loci overlapping expressed
genes. TE → gene: false-positive genes overlapping expressed TEs.' in
'mbcsToSbcs': dot substituted for <86>"
Warning message in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y,
:
"conversion failure on 'Gene → TE: false-positive TE loci overlapping expressed
genes. TE → gene: false-positive genes overlapping expressed TEs.' in
'mbcsToSbcs': dot substituted for <92>"
Warning message in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y,
:
"conversion failure on 'Gene → TE: false-positive TE loci overlapping expressed
genes. TE → gene: false-positive genes overlapping expressed TEs.' in
'mbcsToSbcs': dot substituted for <e2>"
Warning message in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y,
:
"conversion failure on 'Gene → TE: false-positive TE loci overlapping expressed
genes. TE → gene: false-positive genes overlapping expressed TEs.' in
'mbcsToSbcs': dot substituted for <86>"
Warning message in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y,
:
"conversion failure on 'Gene → TE: false-positive TE loci overlapping expressed
genes. TE → gene: false-positive genes overlapping expressed TEs.' in
'mbcsToSbcs': dot substituted for <92>"
Warning message in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x,
x$y, :
"conversion failure on 'Gene-derived reads →' in 'mbcsToSbcs': dot substituted
for <e2>"
Warning message in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x,
x$y, :
"conversion failure on 'Gene-derived reads →' in 'mbcsToSbcs': dot substituted
for <86>"
Warning message in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x,
x$y, :
"conversion failure on 'Gene-derived reads →' in 'mbcsToSbcs': dot substituted
for <92>"
Warning message in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x,
x$y, :
"conversion failure on 'TE-derived reads →' in 'mbcsToSbcs': dot substituted for
<e2>"
Warning message in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x,
x$y, :
"conversion failure on 'TE-derived reads →' in 'mbcsToSbcs': dot substituted for
<86>"
Warning message in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x,
x$y, :
"conversion failure on 'TE-derived reads →' in 'mbcsToSbcs': dot substituted for
```

```
<92>"
Warning message in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x,
x$y, :
"conversion failure on 'Gene → TE: false-positive TE loci overlapping expressed
genes. TE → gene: false-positive genes overlapping expressed TEs.' in
'mbcsToSbcs': dot substituted for <e2>"
Warning message in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x,
x$y, :
"conversion failure on 'Gene → TE: false-positive TE loci overlapping expressed
genes. TE → gene: false-positive genes overlapping expressed TEs.' in
'mbcsToSbcs': dot substituted for <86>"
Warning message in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x,
x$y, :
"conversion failure on 'Gene → TE: false-positive TE loci overlapping expressed
genes. TE → gene: false-positive genes overlapping expressed TEs.' in
'mbcsToSbcs': dot substituted for <92>"
Warning message in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x,
x$y, :
"conversion failure on 'Gene → TE: false-positive TE loci overlapping expressed
genes. TE → gene: false-positive genes overlapping expressed TEs.' in
'mbcsToSbcs': dot substituted for <e2>"
Warning message in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x,
x$y, :
"conversion failure on 'Gene → TE: false-positive TE loci overlapping expressed
genes. TE → gene: false-positive genes overlapping expressed TEs.' in
'mbcsToSbcs': dot substituted for <86>"
Warning message in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x,
x$y, :
"conversion failure on 'Gene → TE: false-positive TE loci overlapping expressed
genes. TE → gene: false-positive genes overlapping expressed TEs.' in
'mbcsToSbcs': dot substituted for <92>"
Warning message in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y,
:
"conversion failure on 'Gene-derived reads →' in 'mbcsToSbcs': dot substituted
for <e2>"
Warning message in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y,
:
"conversion failure on 'Gene-derived reads →' in 'mbcsToSbcs': dot substituted
for <86>"
Warning message in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y,
:
"conversion failure on 'Gene-derived reads →' in 'mbcsToSbcs': dot substituted
for <92>"
Warning message in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y,
:
"conversion failure on 'TE-derived reads →' in 'mbcsToSbcs': dot substituted for
<e2>"
Warning message in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y,
```

```
:
"conversion failure on 'TE-derived reads →' in 'mbcsToSbcs': dot substituted for
<86>"
Warning message in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y,
:
"conversion failure on 'TE-derived reads →' in 'mbcsToSbcs': dot substituted for
<92>"
Warning message in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y,
:
"conversion failure on 'Gene → TE: false-positive TE loci overlapping expressed
genes. TE → gene: false-positive genes overlapping expressed TEs.' in
'mbcsToSbcs': dot substituted for <e2>"
Warning message in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y,
:
"conversion failure on 'Gene → TE: false-positive TE loci overlapping expressed
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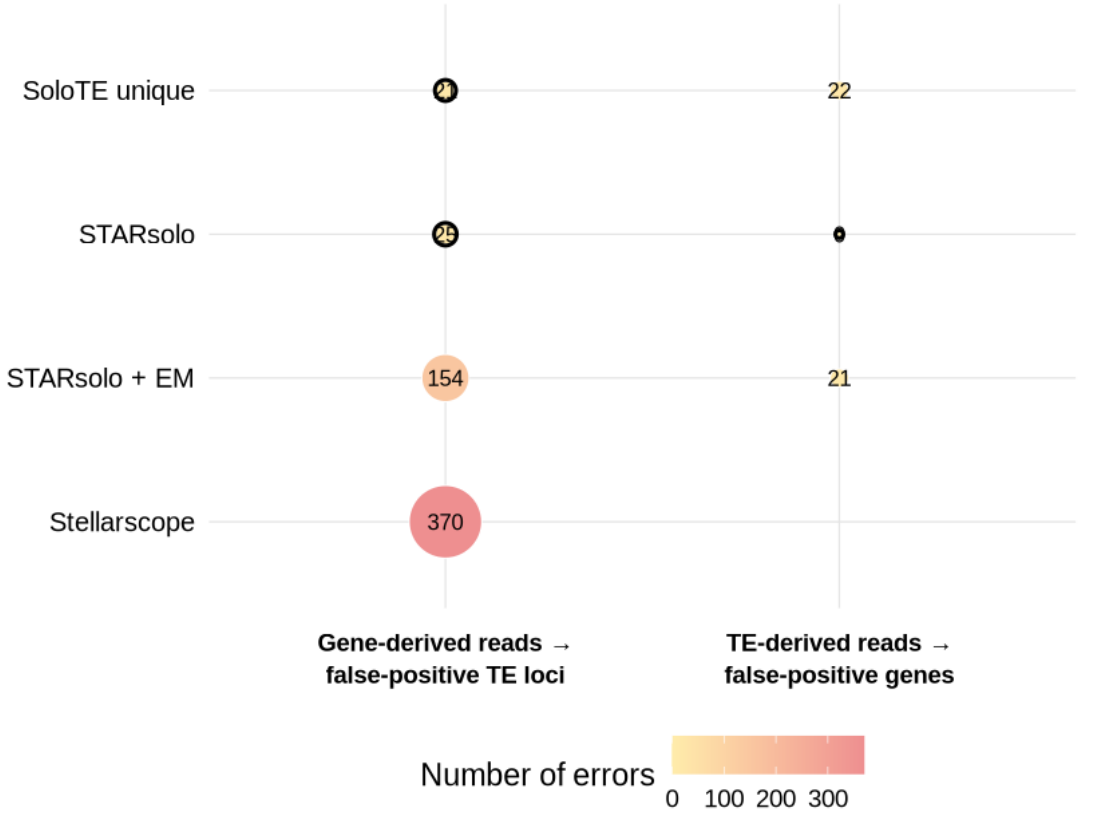

Gene → TE: false-positive TE loci overlapping expressed genes. TE → gene: false-positive
Black outlines indicate methods within five errors of the lowest count in each direction.